

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: 2.1 PROPERTIES OF WAVES – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How do sound waves reflect, and why do some places produce stronger echoes than others? Use what you learned in Lessons 1 to 12 to write your Final Explanation. Answer each section in the space under it. Use the Echo Table and the Surface Table below, and show every calculation with units. Speed of sound in air is 340 m/s. Use the echo-distance formula $d = vt/2$, where t is the total time from the sound leaving to the sound returning and d is the one-way distance. Write in full sentences and use Claim, Evidence and Reasoning where asked.

ECHO TABLE

Site	Place	Distance to nearest large hard surface	What the learners heard
A	Hell's Gate cliff face	85 m	Clear echo
B	Ngong Hills valley wall	170 m	Clear echo
C	Kisumu school hall	12 m	Loud, muddy, drawn-out sound
D	Nairobi school open grounds	300 m (classroom block)	No echo

SURFACE TABLE

Surface	Approximate share of sound energy reflected
Cliff rock	90%
Painted concrete wall	85%
Wooden board	60%
Curtains or cloth	15%
Thick grass or foam mattress	10%

Part 1: What is happening to the sound? (Claim)

Wanjiru shouts towards the cliff at Hell's Gate and hears her voice come back. Describe what the sound wave does from the moment she shouts until she hears the echo. In your answer: (a) say what kind of wave sound is and what it needs to travel; (b) describe the path of the sound, using the idea of a ray; (c) state the two laws of reflection and say how they apply to the sound at the cliff; (d) say what we mean by an echo. Draw a labelled ray diagram showing the normal, the incident ray, the reflected ray, i and r.

Sound is a longitudinal mechanical wave, so it needs a medium; here the medium is air. The shout leaves Wanjiru and travels in a straight line, which we draw as a ray (a straight line with an arrow showing the direction of travel of the wave energy), to the cliff face. At the cliff the sound reflects. The two laws of reflection are: (1) the angle of incidence equals the angle of reflection ($i = r$), both measured from the normal, not from the surface; (2) the incident ray, the reflected ray and the normal all lie in the same plane. These laws work for light, water waves and sound. The reflected sound travels back in a straight line to the listeners. An echo is a distinct reflected sound heard after the original sound, at least 0.1 s later. Diagram: a surface line (the cliff) with a dashed normal at right angles to it; an incident ray arriving at the normal point with angle i marked from the normal; a reflected ray leaving on the other side with angle r marked from the normal and equal to i ; arrows on both rays.

Part 2: Delay and distance (Evidence)

Use the echo-distance formula to work out the following. (a) Show that the delay for the echo at Hell's Gate (site A) is 0.50 s. (b) Calculate the delay for Ngong Hills (site B). (c) Calculate the delay for the Kisumu hall (site C) and the one for the open Nairobi grounds (site D). (d) Explain why the ear needs a delay of at least 0.1 s to hear a distinct echo, and calculate the smallest distance to a reflector that gives a distinct echo. (e) Kiprono stands in front of a long school wall and times 10 clap-echo intervals as 2.94 s. Find the distance to the wall.

The delay is $t = 2d/v$ because the sound travels to the surface and back. (a) Site A: $t = 2 \times 85 / 340 = 170 / 340 = 0.50$ s. (b) Site B: $t = 2 \times 170 / 340 = 340 / 340 = 1.00$ s. (c) Site C: $t = 2 \times 12 / 340 = 24 / 340 = 0.07$ s. Site D: $t = 2 \times 300 / 340 = 600 / 340 = 1.76$ s. (d) The ear can separate two sounds only if they are at least 0.1 s apart. The smallest distance is $d = vt/2 = 340 \times 0.1 / 2 = 17$ m. Sites A, B and D are more than 17 m from their reflectors, while site C (12 m, 0.07 s) is less than 17 m, so its reflection arrives too soon to be heard as a separate echo. (e) One interval = $2.94 / 10 = 0.294$ s. $d = vt/2 = 340 \times 0.294 / 2 = 49.98$ m, which is about 50 m.

Part 3: Strong echoes and weak echoes (Reasoning)

Delay and distance alone do not explain every site. Use the Echo Table and the Surface Table to answer: (a) Why do Hell's Gate and Ngong Hills give clear, strong echoes? (b) Why is the sound in the Kisumu hall loud, muddy and drawn out? Use the words reverberation and interference in your answer. (c) Site D has a delay of 1.76 s, which is longer than 0.1 s, yet there is no echo. Explain why, using the ideas of spreading (diffraction) and absorption. (d) Which surface in the Surface Table would give the strongest echo, and which the weakest? Give a reason.

(a) At Hell's Gate (85 m) and Ngong Hills (170 m) the surfaces are far more than 17 m away, so the delays (0.50 s and 1.00 s) are longer than 0.1 s and the echo is heard separately from the shout. The cliff and valley wall are large, hard, smooth rock surfaces, which reflect about 90% of the sound energy, so the returning sound is strong. (b) In the Kisumu hall the far wall is only 12 m away, so the delay is 0.07 s, less than 0.1 s, and the reflection merges with the voice. The hall also has many hard walls and a hard roof, so the sound keeps reflecting and the reflections overlap. This overlapping and prolonging of sound is reverberation, and it lasts about 1.2 s. The overlapping reflected waves add together (interference): in some places they strengthen the sound and in others they partly cancel it, so the sound is loud, muddy and drawn out. (c) On the open grounds the sound spreads out over a huge area (it diffracts and spreads), so only a small part reaches the classroom block 300 m away. The grass and bushes also absorb and scatter much of the energy. The 1.76 s delay is long enough, but the returning sound is too faint to hear, so there is no echo. A strong echo needs both a distance of at least 17 m and a surface that reflects enough energy. (d) Cliff rock (90%) gives the strongest echo because it reflects most of the energy. Thick grass or foam (10%) gives the weakest because it absorbs most of the energy.

Part 4: Other wave behaviours at a boundary

When a wave meets a boundary, different things can happen. (a) Explain the difference between reflection and refraction, and say which of them returns the sound to the learners at Hell's Gate. (b) Light going from air into water at $i = 45$ degrees bends to $r = 32$ degrees. State what this tells you about the speed of light in water and why the light bends. (c) Sound can turn the corner of a 1.0 m doorway but a torch beam makes a sharp patch. Use the wavelength of a 340 Hz sound and the wavelength of light (5.5×10^{-7} m) to explain this.

(a) In reflection the wave bounces back into the same medium with $i = r$. In refraction the wave passes into a second medium and changes direction because its speed changes. At Hell's Gate it is reflection that sends the sound back and makes the echo; refraction does not by itself explain the echo. (b) The light bends towards the normal, so it slows down in water. Light travels at 3.0×10^8 m/s in air and 2.26×10^8 m/s in water. The wave changes direction because its speed changes at the boundary. (c) Wavelength = $v / f = 340 / 340 = 1.0$ m for a 340 Hz sound. This is about the same size as the 1.0 m doorway, so the sound spreads strongly (diffracts) and can be heard around the corner. The wavelength of light, 5.5×10^{-7} m, is tiny compared with the doorway, so the light hardly bends and makes a sharp patch with a clear edge. Diffraction does not change the frequency or the wavelength.

Part 5: Final Explanation and applying the idea

Write your Final Explanation of the driving question: How do sound waves reflect, and why do some places produce stronger echoes than others? Use Claim, Evidence and Reasoning. Your answer must cover all four sites (A to D) and use at least three numbers from the tables. Then answer: (a) Mr. Mwangi asks you to reduce the reverberation in the Kisumu hall. Suggest a treatment from the Surface Table and explain how it works. (b) Give one way the echo idea is used in medicine and calculate the time for an ultrasound pulse (1540 m/s) to return from an organ boundary 7.7 cm deep.

Claim: Sound is a mechanical wave that travels from the shouting learner, reflects from a large hard smooth surface following the law of reflection ($i = r$), and returns after a delay $t = 2d/v$. A strong separate echo needs a reflector at least 17 m away that reflects enough energy. Evidence: At Hell's Gate (A) the cliff is 85 m away, giving a delay of 0.50 s, and at Ngong Hills (B) the wall is 170 m away, giving 1.00 s; both are hard surfaces (rock reflects about 90%), so the echoes are clear. In the Kisumu hall (C) the wall is 12 m away, a delay of 0.07 s, less than 0.1 s. On the open Nairobi grounds (D) the delay is 1.76 s but there is no echo. Reasoning: At A and B the distance is more than 17 m and the surface is hard, so a strong echo returns separately. In the hall the reflection arrives too soon to be separate, and the many hard surfaces keep reflecting, so the overlapping reflections interfere and cause reverberation of about 1.2 s: loud, muddy and drawn out. On the open grounds the sound spreads out and the grass absorbs it (grass reflects only about 10%), and the only big wall is 300 m away, so the returning sound is too faint to hear. (a) Hang curtains (15% reflected) or put foam (10%) on the walls and roof. These materials absorb sound energy instead of reflecting it, so fewer reflections overlap and the reverberation is shorter. (b) Ultrasound scanners send pulses that reflect from boundaries between tissues, and the echo time shows the depth using $d = vt/2$. $t = 2d/v = 2 \times 0.077 / 1540 = 1.0 \times 10^{-4}$ s, which is 100 microseconds.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim: describing reflection and the laws of reflection	Clearly states that sound is a mechanical wave that travels as a ray, reflects at the surface and returns; states both laws of reflection correctly with angles measured from the normal; the ray diagram is fully labelled and accurate.	States that sound reflects and returns and gives at least one law of reflection correctly; the diagram has most labels but may miss the normal or one angle.	Says sound bounces back but does not state the laws, or measures angles from the surface; the diagram is missing or unclear.
Evidence: calculations with $d = vt/2$ and the 17 m condition	All delays (0.50 s, 1.00 s, 0.07 s, 1.76 s), the 17 m minimum distance and the 50 m field distance are correct, with working and units shown.	Most calculations are correct with the right formula, but there are small slips in arithmetic or missing units.	Uses the formula incorrectly (for example forgetting to halve or double) or gives several wrong values.
Reasoning: explaining the four sites using surfaces, reverberation, spreading and absorption	Explains all four sites correctly, linking distance, delay, surface reflection percentages, reverberation and interference in the hall, and spreading and absorption on the grounds.	Explains most sites correctly but leaves out one idea, such as absorption or the overlapping reflections.	Gives only descriptions of what was heard, or explains with ideas not supported by the data.
Connecting other wave behaviours and applications	Correctly separates reflection from refraction, explains bending using speed	Gets most of the distinctions right and does most of the wavelength or	Confuses reflection with refraction or does not use wavelength to explain

	change, uses wavelengths to explain diffraction, and calculates the ultrasound echo time of 100 microseconds.	ultrasound working correctly.	diffraction.
Communication using Claim, Evidence and Reasoning	The Final Explanation has a clear claim, uses at least three numbers from the tables as evidence, covers all four sites, and uses scientific terms correctly and consistently.	Has a claim and some evidence, covers three or four sites, and mostly uses correct terms.	Claim, evidence and reasoning are mixed together or missing, and few data are used.

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